



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO GUBER AGENCY IOC DEALERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

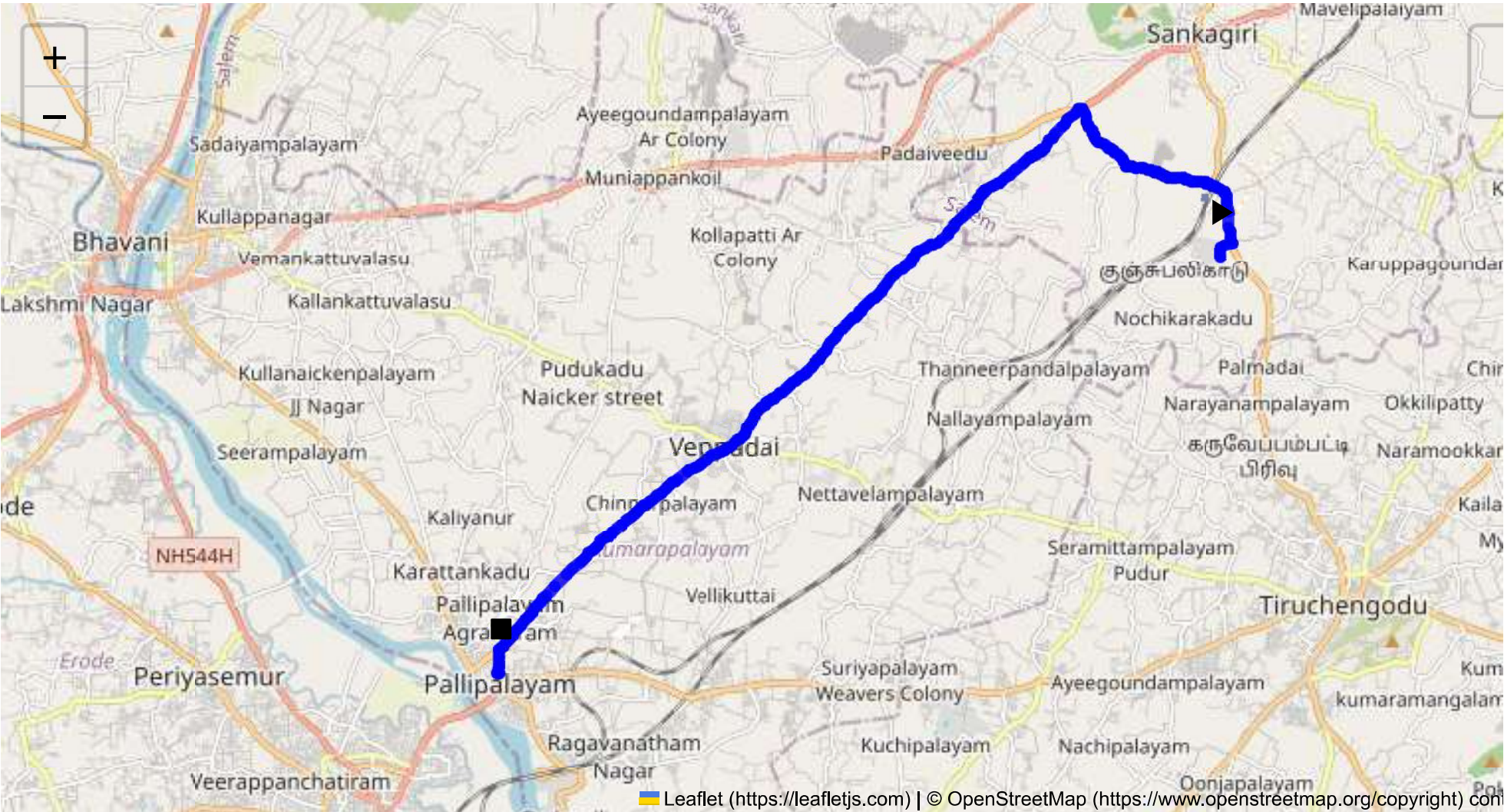
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 20.47 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.366803, 77.748303)



Welcome to the Journey Risk Management Study

Route Analysis: CVQF+23W, Sangagiri to 14, Pallipalayam - Sankari Rd, Tamil Nadu

1. Overview of the Route Map

The route from Sangagiri to Pallipalayam is approximately 20.47 kilometers long, typically taking about 27-30 minutes for heavy vehicles. The route primarily follows state highways and local roads, traversing through a mix of rural and semi-urban areas.

2. Typical Weather Conditions and Hazards

Tamil Nadu experiences tropical weather, characterized by a hot and humid climate for most of the year. Monsoon season, from June to September, can bring heavy rainfall, leading to potential road flooding and reduced visibility. During summer months, temperatures are high, leading to the risk of tire blowouts or engine overheating in heavy vehicles.

3. Traffic Patterns

Peak traffic hours are usually from 8:00-10:00 AM and 5:00-7:00 PM. Congestion is most likely to occur around entry and exit points of towns or near market areas. Being mostly a rural and semi-urban route, traffic is moderate but can increase during local festivals or market days.

4. Road Quality and Infrastructure

The state highway is generally well-maintained with clear signage, but local roads may present issues such as potholes or lack of proper markings. Ensure thorough inspections of vehicle suspension and tires to manage wear and tear effectively.

5. Suggestions for Alternative Routes

For emergencies, routes via Tiruchengode may be considered; however, alternative routes should be verified for road conditions, as they can vary significantly.

6. Local Regulations on Hazardous Material

Tamil Nadu’s transport regulations require permits for hazardous material transport. Restrictions may apply in densely populated or sensitive areas, and compliance with all safety protocols is mandatory.

7. Overview of Historical Incidents

While specific data on this particular route may not be readily available, incidents involving heavy vehicles typically relate to mechanical failures, weather-related slip-ups, or human error, particularly during adverse weather.

8. Environmental Considerations

The region includes agricultural areas that require cautious driving to prevent contamination. Non-compliance with environmental norms can attract hefty fines, especially near water bodies or ecologically sensitive zones.

9. Communication Coverage

Communication coverage is generally reliable along state highways; however, some rural stretches may experience occasional dead zones. Drivers should be equipped with emergency communication devices.

10. Estimated Emergency Response Times

Emergency services in rural and semi-urban areas may take 30-45 minutes to reach due to distance from major centers. Familiarizing with locations of local hospitals and fire stations is advised.

12. Overall Risk Assessment

Risk on this route is moderate. Potential hazards include weather during monsoon, road conditions on semi-urban stretches, and moderate traffic congestion. Planning trips outside peak hours, ensuring vehicle maintenance, and awareness of local regulations will mitigate most risks.

Conclusion

This analysis identifies important considerations for safer transport along the route between Sangagiri and Pallipalayam. Awareness of weather conditions, traffic patterns, and adherence to regulations will help minimize risks. Regular updates of route conditions and local advisories are recommended for ongoing safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.37086, 77.74828	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.46307, 77.84941	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.46312, 77.84886	30 KM/Hr
8	Turn	High	11.36696, 77.74794	15 KM/Hr
9	Turn	Medium	11.36692, 77.74807	30 KM/Hr
10	Turn	High	11.36690, 77.74809	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Cheran Hospital	11.3680302, 77.7481976	30 km/h	Medium
2	hospital	Sri Kumaran E N T Hospital	11.3657814, 77.7477763	30 km/h	Medium
3	pharmacy	Pharmacy	11.3652161, 77.7486375	30 km/h	Medium
4	hospital	Sri Krishna Hospital	11.3656648, 77.7477783	30 km/h	Medium
5	hospital	Visaalam Hospitals	11.3651528, 77.747808	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Ssri Valliappa Vidhayalayam mat hr school	11.4004878, 77.7794381	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37086, 77.74828



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36696, 77.74794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36692, 77.74807



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36690, 77.74809

